Solar Filament Instance Segmentation on GONG Hα: a P2/stride-2 YOLO detector, crop refiner and agreement-calibrated keep rule

[TODO: team name / authors] Repository: [TODO: public repo URL]

Abstract

We segment solar filaments on 2048^2 GONG Hα images with a P2-augmented YOLO11s-seg detector, 4-flip consensus, a ResNet18 U-Net crop refiner and a learned keep rule that demotes masks a random annotator is unlikely to have drawn. On 5-fold out-of-fold (OOF) validation over all 707 training files the final pipeline reaches PQ **0.4590** (public LB 0.39). Human annotators agree with each other at only PQ ≈0.34, and most residual error is annotator disagreement rather than mask quality.

1 Task, metric and validation

Data. 707 training files with 1154 annotator records and 8199 polygons (411 files have one record, 145 two, 151 three); 180 test images. **Metric.** Panoptic Quality with IoU>0.5 matching, where each prediction set is matched separately against every annotator record and TP/FP/FN are pooled over all (image, record) pairs. A filament drawn by k of n annotators and predicted with IoU s thus earns k TP and $n-k$ FP, so predicting it pays only if roughly $k/n > \mathrm{PQ}/(2s) \approx 0.31$: the target is "would a random annotator draw this", not "is this a filament". Our scorer reproduces the host self-evaluation notebook exactly (identical TP/FP/FN on all 707 files).

Validation protocol. 5-fold OOF split by file (141–142 files per fold); every file is predicted by detector, refiner and keep-rule models that never saw it (the refiner and the keep rule are refit per fold: "leak-free" refined OOF). Each experiment is *pre-registered*: arms, metric, and a mechanical CONTINUE/KILL gate are written and hashed (sha256) before any number exists. Comparisons use a paired file-level bootstrap (2000 resamples) and report ΔPQ, 95% CI, $P(\Delta > 0)$ and per-fold signs. Typical gate: pooled ΔPQ≥+0.010 (screens) or ≥+0.005 (final bundle), $P \geq 0.90$, ≥4/5 folds positive; KILL if ΔPQ<+0.005. Seed/retraining noise is measured, not assumed (detector seed noise −0.005 on fold 0; refiner retraining noise ≈0.0001). The public LB shows two decimals on ∼90 images and cannot resolve the effects below, so it was not used for selection.

2 Pipeline

(1) Detector: "p2first" YOLO11s-seg. We add a P2 level to YOLO11s-seg (Upsample → Concat with backbone P2 → C3k2[128]) and place it *first* in the Segment head inputs [P2,P3,P4,P5]. Because Ultralytics builds mask prototypes from the first input, this gives stride-2 prototypes (512×512 at 1024 input) plus stride-4 detection, which matters for thin, small filaments. Init: COCO YOLO11s-seg with head tensors re-indexed (509/639 tensors transfer). Training: single class, imgsz 1024, batch 4, 80 epochs at mask_ratio 2 then 20 epochs at mask_ratio 1 (AdamW 10^{-4}); one record = one training image. The final detector is retrained on all 707 files.

Table 1: 5-fold OOF, 707 files, pooled per-record PQ. Baseline = V65 deployed path (p2first + r3 refiner, $t0.45/c0.30$) with leak-free 5-fold refiner. Paired bootstrap vs baseline.

pipeline	PQ	SQ	TP/FP/FN	ΔPQ [95% CI]
V65 path (baseline)	0.4539	.685	5270/2444/2929	–
V84 (final)	**0.4590**	.688	5050/1881/3149	**+0.0050** [+.0024,+.0077]
only soft targets	0.4550	.685		+0.0011 [+.0001,+.0021]
only $t0.50$	0.4546	.686		+0.0007 [−.0001,+.0015]
only MIN200	0.4560	.685		+0.0021 [+.0014,+.0029]
only demote	0.4569	.687		+0.0030 [+.0007,+.0053]
V84 − MIN200	0.4590	.688		+0.0051 [+.0023,+.0077]
V84 − demote	0.4567	.686		+0.0028 [+.0013,+.0044]

(2) 4-flip TTA and consensus. Inference on identity, H, V and HV flips; masks are clustered across views (anchor = highest score, IoU 0.5), cluster score = sum of view scores / 4. **(3) Pixel ownership.** Score-ordered ownership makes instances pixel-disjoint (required by the host).

(4) Crop refiner. A ResNet18 U-Net on 256-px proposal-centred crops (4-flip TTA) re-predicts each mask at full 2048 resolution, clamped to a 7×7 dilation of its seed. It is trained on the detector's *own OOF proposals* in two stages: (A) 8 epochs on GT crops (lr $3 \cdot 10^{-4}$), (B) 12 epochs on proposals matched to GT (IoU≥0.3), 0.5 BCE + 0.5 Dice. V84 uses *soft targets* in stage B: the target is the per-pixel mean, over the R annotator records, of the GT mask each record matched to the proposal (0 for records without a match), so the refiner learns agreement-weighted extent. Mask threshold $t = 0.50$, detection cut $c = 0.30$.

(5) MIN200 + demote rule. Masks below 200 px are dropped. A LightGBM regressor (28 features: consensus and member scores, view support, ownership retention, refined-probability statistics, area, elongation, width, disk position, contrast, disk statistics, neighbour and per-file counts) predicts for each kept mask j the expected per-record matched quality $\hat{y}_j \approx q_j s_j$, where q_j is the fraction of records that match it and s_j the mean matched IoU. Since $\mathrm{PQ} = \sum qs/(\mathrm{TP} + \frac{1}{2}\mathrm{FP} + \frac{1}{2}\mathrm{FN})$, adding mask j raises PQ iff $q_j s_j > \mathrm{PQ} \cdot \frac{1}{2}$; the optimal keep set is therefore a threshold on $\hat{y}$ at $\lambda/2$, with λ the fixed point $\lambda = \mathrm{PQ}(\{j : \hat{y}_j > \lambda/2\})$ found by Dinkelbach iteration (converges in a few steps; $\lambda = 0.4587$ on pooled cross-fitted OOF). Rule: keep j iff $\hat{y}_j > \lambda/2$. Only demotion is used (promoting masks below c gave +0.0001). On OOF it removes 9.3% of masks (4686→4248), on test 8.3%.

(6) Export. Native 2048 masks, COCO RLE, validated (non-overlap, full decode) before writing.

Table 2: Selected pre-registered negatives Direct = detector only; refined = through the refiner. 141 held-out files (fold-0 screens or canonical hold-out), except refiner variants and re-scorer (707-file 5-fold).

change	direct ΔPQ	refined ΔPQ [95% CI]
imgsz 1280	+0.0143	−0.0029 [−.013,+.007]
thin labels (−1 px polygons)	+0.0272	−0.0017 [−.010,+.007]
noisy student (unlab. GONG)	+0.0106	+0.0007 [−.010,+.011]
label transfer ±1/2/5 min	−.011/+.001/−.006	not run (killed)
2-detector ensemble	+0.0063	+0.0016 [−.006,+.009]
refiner: union targets	–	−0.0195 [−.026,−.013]
refiner: quality-weighted	–	−0.0000 [−.002,+.002]
refiner: crop512 / RN34 / dil15	–	+.0000 / +.0006 / +.0006
learned re-scorer (agreement)	+0.0161	−0.0099 [−.018,−.002]

3 Results

Table 1 gives the final comparison. The V84 bundle gains ΔPQ = +0.0050, CI [+0.0024, +0.0077], $P(\Delta > 0) = 0.9995$, positive on 5/5 folds (+.0057, +.0097, +.0028, +.0027, +.0045). It met its pre-registered +0.005 bar by only 0.00004, so we expect ≈+0.005±0.003 on test, not a certain gain. Single components sum to +0.0069; the bundle keeps 73%. The demote rule is the largest marginal component and makes MIN200 redundant (the bundle without MIN200 is +0.00002 better; MIN200 is kept as a harmless export default). Both V65 and V84 score **0.39** on the public LB, which is consistent with a +0.005 effect being below its two-decimal resolution.

Earlier milestones (direct detector, 707 files): baseline YOLO11s-seg 0.3856 → p2first 0.4060 (ΔPQ +0.0204 [+0.0151, +0.0256], 5/5 folds; small-object recall 0.411→0.460). The crop refiner adds ≈+0.04–0.06 PQ on top of any detector, mostly via SQ (0.657→0.689).

Negative results: the refiner absorbs detector gains. Table 2 lists pre-registered experiments that were killed by their gates. A recurring finding is that improvements that are clear on the direct detector vanish after refinement: the refiner already fixes mask geometry, so detector gains must come from *instance selection*, not extent. From V76 on, every detector change had to pass a refined-pipeline gate.

Other closed lines: Mask R-CNN, Mask2Former, DenseCRF, BCE+Focal+Dice loss, YOLO11m, size-gated refiner (−0.014), 1-px erosion of refined masks (−0.024). Soft targets (+0.0019, fold 0) and the keep rule (+0.0030, 707 files) failed their *individual* gates and were only adopted as a re-registered bundle.

4 Host evaluation plots

Figure 1 shows the three required distributions on OOF predictions. Over all overlapping pairs the mean IoU is 0.613 (Dice 0.742); over matched pairs (IoU>0.5) 0.688 (Dice 0.812). The IoU mass sits at 0.65–0.80 and drops sharply above 0.85: annotators' own boundaries disagree at that level (human SQ ≈0.63). Topology errors are rare: 134 of 8199 GT (1.6%) overlap ≥2 predictions (fragmentation) and 187 prediction-record pairs (2.7%) overlap ≥2 GT (over-merging); 1:1 dominates on both sides. The 2004 GT with no overlapping prediction are mostly filaments drawn by a single annotator (below).

5 Annotator-disagreement ceiling

On the 296 multi-record training files we scored annotators against each other and computed label oracles (all with access to GT, so they are ceilings, not achievable predictors):

- **Human vs. held-out human:** PQ **0.340** (SQ 0.634; 1196 ordered pairs). Our direct detector scores 0.411 on the same records, so the model already beats a held-out annotator.
- **Oracle predictors** (submit GT masks, scored against all records of the file): union of annotators **0.663**; one annotator's own masks **0.605** (best/worst 0.631/0.571); majority/$k \geq 2$ **0.592**; Dinkelbach-optimal subset of vote masks 0.668. Even perfect masks cap PQ near 0.6–0.67 because records disagree on *which* filaments exist.
- **Keep-oracle:** choosing which of our own existing refined masks to keep using the true $q \cdot s$ gives 0.5283 (**+0.074**); our learned rule realises +0.003 of it. Agreement is weakly predictable from image or refiner evidence (Spearman with true qs: 0.59 vs 0.57 for the score alone).
- Recall on filaments drawn by ≥2 annotators is 0.834 vs 0.367 for single-annotator ones; fixing every missed agreed filament would add only +0.030.

The remaining gap is dominated by label disagreement, which no single-image model can resolve.

6 Data use and rule compliance

- **Training data:** only the provided training images and labels. No external ground truth of any kind; no test labels or pseudo-labels on test images. Pre-trained weights: COCO YOLO11s-seg (Ultralytics), ImageNet ResNet18 (torchvision).
- **Inference:** single-image; each test Hα image alone (no neighbouring frames, no magnetograms).
- **Training-only experiments not in the final pipeline**, run under host rulings: (i) label transfer of training polygons to nearby GONG frames ±1–5 min, excluding any frame within ±1 h of a test timestamp (Kaggle discussion thread 748045); (ii) noisy-student pseudo-labelling on unlabelled GONG Hα frames from days excluding all train and test dates ±1 day (thread 744047). Both were killed by their pre-registered gates (Table 2) and neither data source is used by the submitted models.

7 Compute and efficiency

Training on 4× NVIDIA T4 (one model per GPU; DDP is incompatible with the in-process stride-2 patch). Final all-data detector: 2.4 h (stage 1, 80 ep) + 2.1 h (stage 2, 20 ep) = 4.5 GPU-h; all-data refiner 24 min; LightGBM demote model: seconds on CPU. The full 5-fold OOF protocol costs ≈5× this. **Inference:** 180 test images in 3 shards of 60 on one T4 each, 148–152 s per shard, i.e. **≈2.5 s/image** end-to-end (4-flip detector + 4-flip crop refiner + keep rule, model loading included). [TODO: Kaggle-notebook GPU timing if the host requires it.]

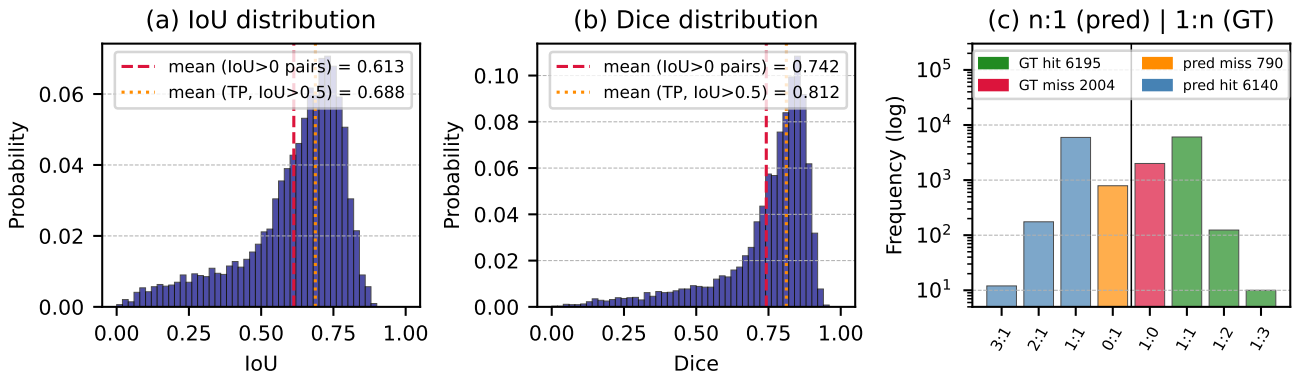


Figure 1: Host self-evaluation plots for the V84 pipeline on 5-fold OOF predictions (707 training files, 8199 GT polygons over all records; PQ 0.4590 reproduced). Logic mirrors the host notebook (`plot_distribution`, `plot_m2n_counts`): all GT–prediction pairs with IoU>0 per annotator record. (a) IoU, (b) Dice (=2IoU/(1+IoU) per pair), (c) number of overlapping partners per GT (1:n, right) and per prediction (n:1, left).

8 Limitations

The final gain over V65 (+0.005) is small and passed its gate narrowly; the all-data refiner and demote model are single fits. Thin structures (barbs; agreed filaments narrower than 5 px: recall 0.60) remain the hardest morphology. The largest open lever is predicting inter-annotator agreement (+0.074 keep-oracle headroom), which our features could not reach. [TODO: qualitative Hα overlay figure (OOF, refined vs. $\geq$2 annotators) if space allows.]